*57% of adults experienced a scam in the past year, with 23% reporting money stolen*

— Forbes (quoting Google?)

# Session 9: Safety

## Tech Skills 101: Driver's Ed for the Digital World

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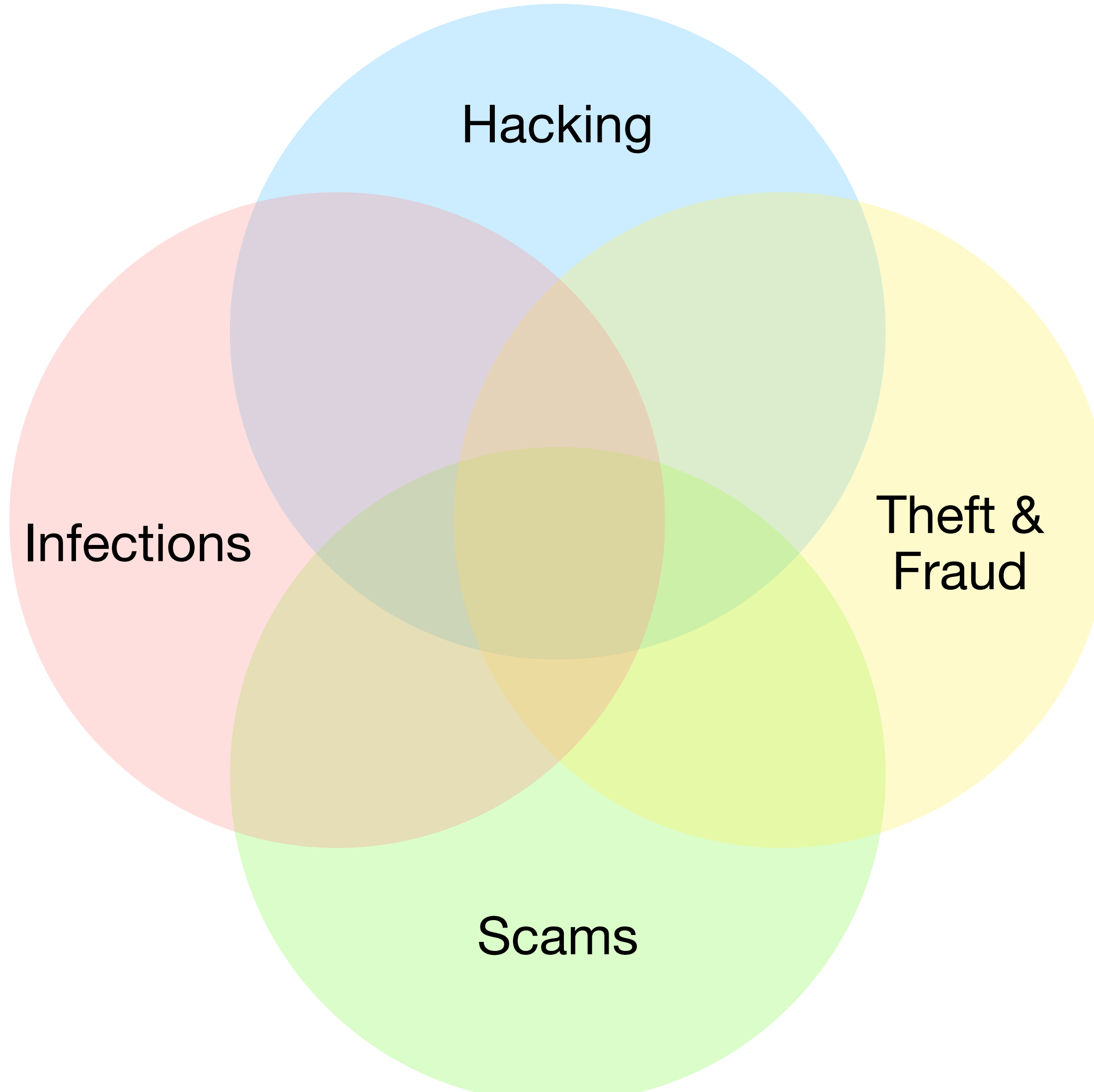
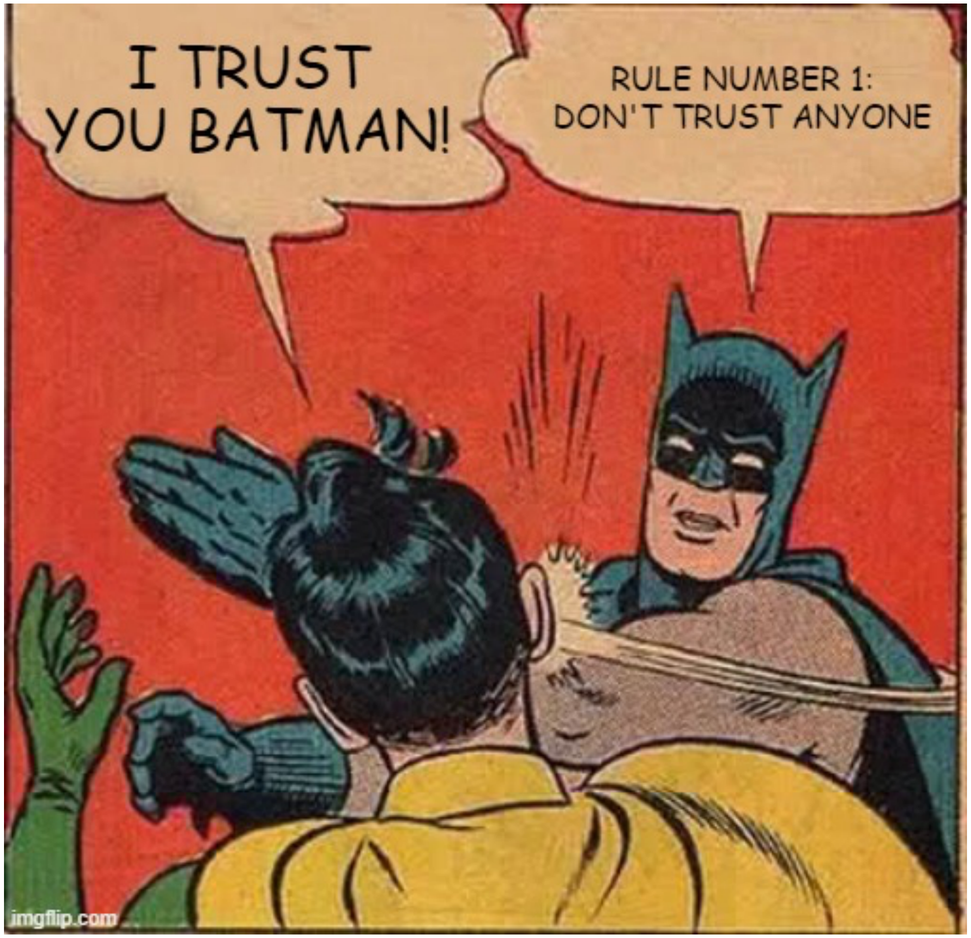
**OLLI Fall 2025**

# “Big Picture” Context

- Safety
  - Security (a.k.a. cybersecurity): Avoiding getting “hacked”, having data stolen, falling victim to scams and/or getting viruses
  - Well-being (i.e., mental health): Use of some sites can contribute to/cause mental health issues (e.g., addiction, body dysmorphia, suicidal ideation/ tendencies)
- Privacy: What I do and how I do it can reveal aspects of my personal life to others

# Security Risks




As we rely more on digital platforms for everything from banking to communication, the concept of trust in cybersecurity takes center stage. ***Trust here means believing that our online interactions and data are safe from cyber threats***. Yet, assuming this trust without verification can lead to significant vulnerabilities.

It's important to understand that cybersecurity is not just about setting up defenses but ***continuously ensuring they are effective against evolving threats***.

Traditional trust assumptions in cybersecurity fall short due to the rapidly evolving threat landscape, where attackers continually develop new methods to exploit vulnerabilities. Misplaced trust in systems or individuals without proper verification often leads to breaches, highlighting the urgent need for adaptive and robust cybersecurity strategies.

*Ref: national1927.com*

# Security Best Practices

- Use **strong** passwords
- Use **unique** passwords
- Use **two-factor authentication** (2FA)
  a.k.a. multi-factor authentication (MFA)
- Routinely change your passwords
- Use **passkeys**
- Use a **password manager**
- Use **VPN**
- Use **antivirus** software
- **Encrypt** *confidential* data you store *in the cloud*

★ Keep software (apps & OS) **up to date**

These items relate to protecting your online accounts.

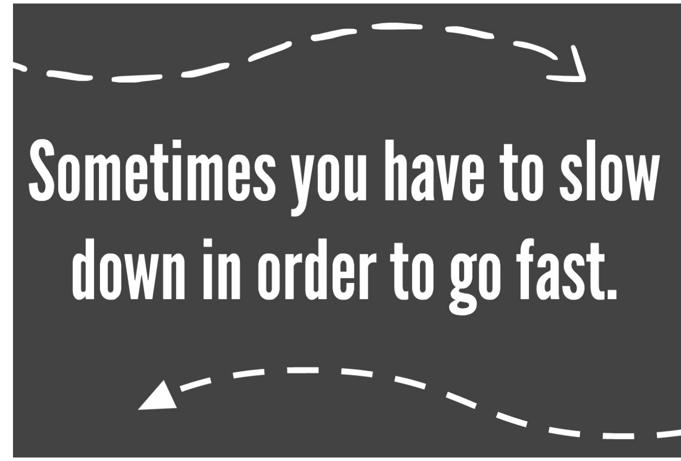
A virtual private network (VPN) protects you when you use "public" WiFi networks.

Antivirus software scans programs and files as they enter your device and compares them to known viruses, or it scans programs already on your device, looking for any suspicious behavior. Modern antivirus solutions do both. In addition, most antivirus software features tools to either remove or quarantine offending malware.

If you must save confidential data on your cloud storage, then make sure the cloud storage you choose to us supports encryption of "data at rest" and that you turn it on.

Apple iCloud and Microsoft OneDrive do support "data at rest" encryption but it is not turned on by default. Google's cloud has encryption by default **and** Google owns the keys (meaning your control is limited; Google can read your content; if Google is hacked, the hackers could get access to your content.

# The Hard Truth



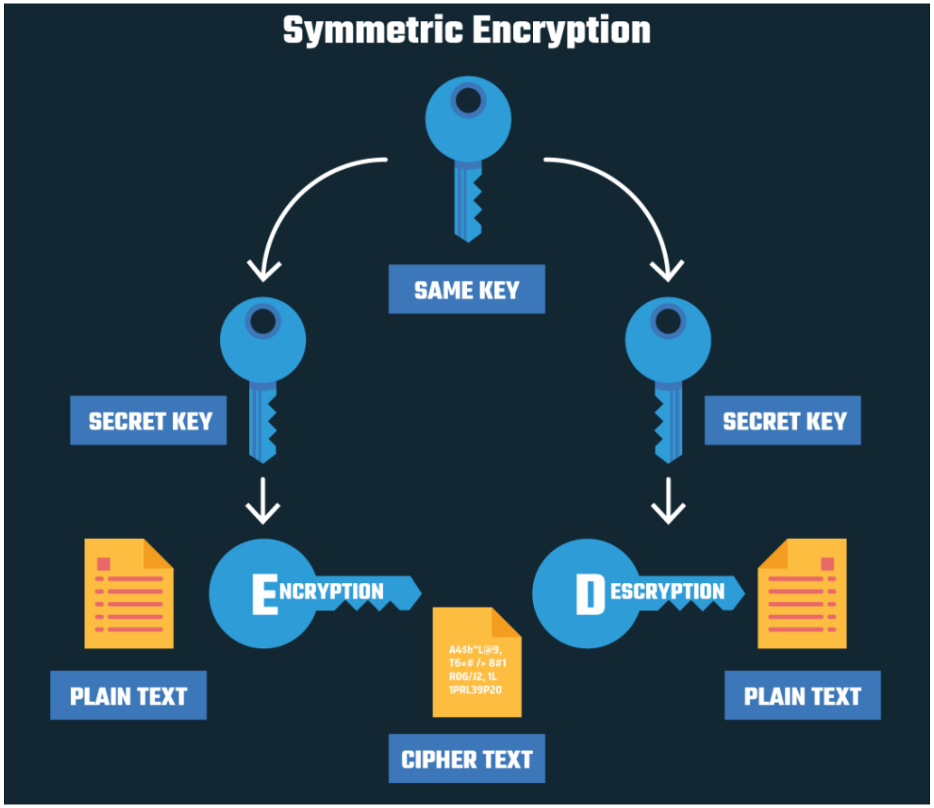

- The bad news
  - There is no bulletproof way to avoid the “bad guys”
  - There are ways to reduce the risk of falling victim to the “bad guys”
  - Taking proactive/defensive steps also helps protect others
- The good news
  - You don’t necessarily need to follow every best practice recommendation
    Pick the items that make the most sense to you and pursue them a step at a time
  - Start small; take it one step at a time — each step reduces your risk

# Encryption: Defintion

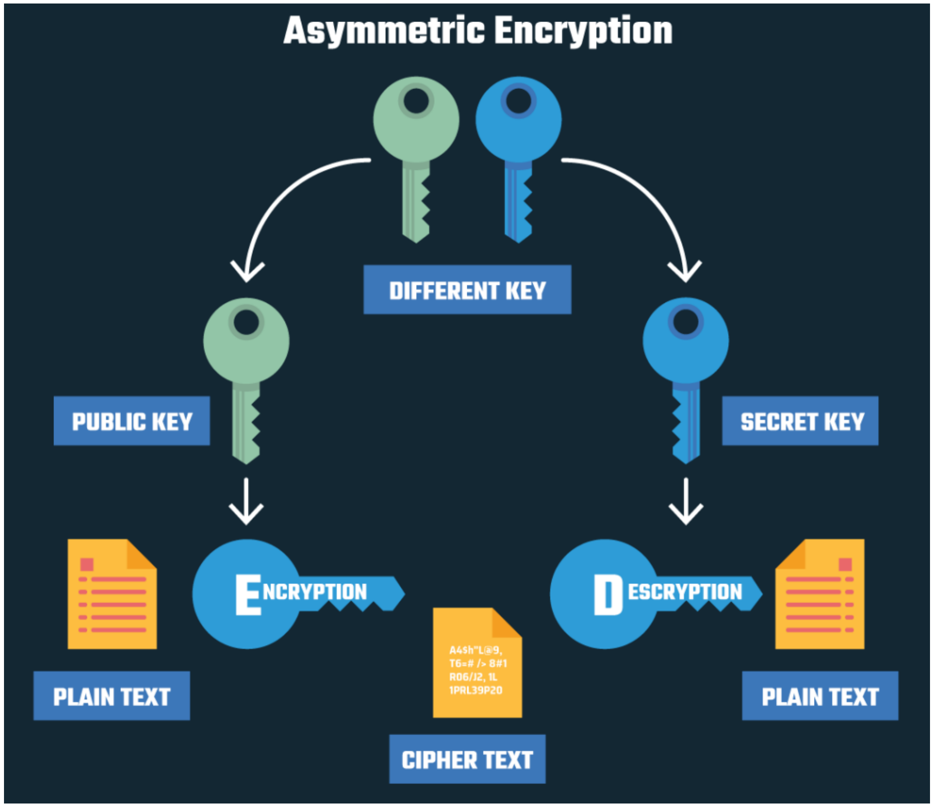
- **Encryption** converts sensitive information or data into a secret code to prevent unauthorized access.
- Encryption scrambles plain text into a type of secret code that hackers, cybercriminals, and other online snoops can't read—even if they intercept it before it reaches its intended recipients. When the message gets to its recipients, they have their own key to unscramble the information back into plain, readable text.
- Encryption takes plain text, like a text message or email, and scrambles it into an unreadable format called ciphertext. This helps protect the confidentiality of digital data either stored on computer systems or transmitted through a network like the internet.


- When the intended recipient accesses the message, the information is translated back to its original form. This is called **decryption**.
- To unlock the message, both the sender and the recipient have to use a “secret” **encryption key**—a collection of algorithms that scramble and unscramble data back to a readable format.

# Encryption: Types




Symmetric encryption is used to protect large amounts of data with low latency (i.e., encryption/decryption is fast).




Asymmetric encryption is used to assert trust, assure identify, and to digitally sign documents/content.

# Encryption: Common Algorithms

**Symmetric**

- Triple DES (3DES)
- Advanced Encryption Standard (AES)
- Rivest-Shamir-Aldeman (RSA)
- RC5 and RC6
- Twofish
- Elliptic Curve Cryptography (EEC)

**Asymmetric**

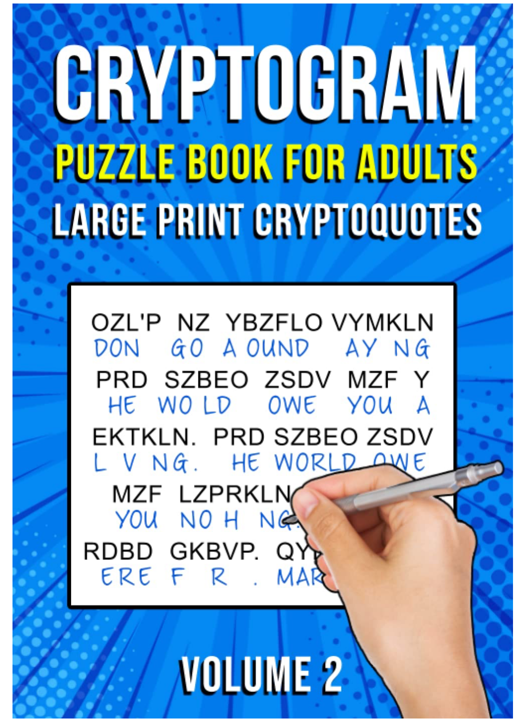
- Rivest-Shamir-Adleman (RSA)
- Elliptic Curve Cryptography (EEC)
- Diffie-Hellman
- DSS (Digital Signature Standard)
- El Gamal
- PGP and S/MIME

# Encryption: Strength

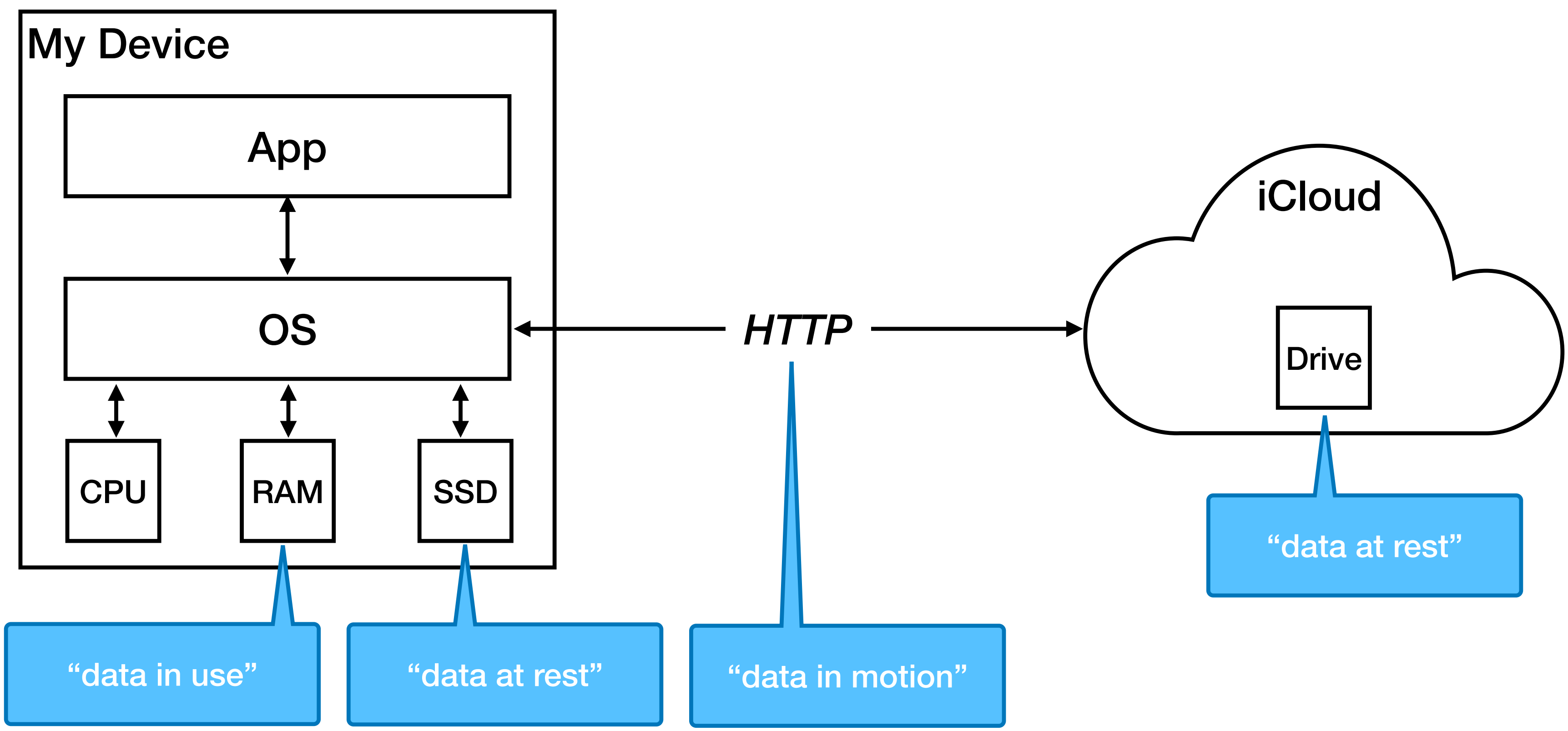
- **Strong encryption** refers to encryption algorithms that, when used correctly, provide a very high (usually insurmountable) level of protection.
- An encryption algorithm's strength is usually defined as a number of bits used for the key.
- Currently a minimum of **128 bits** for keys is considered the **minimum** length for a strong encryption. It is quite common these days to encounter encryption algorithms that use 256 bits.

# Encryption: Cracking

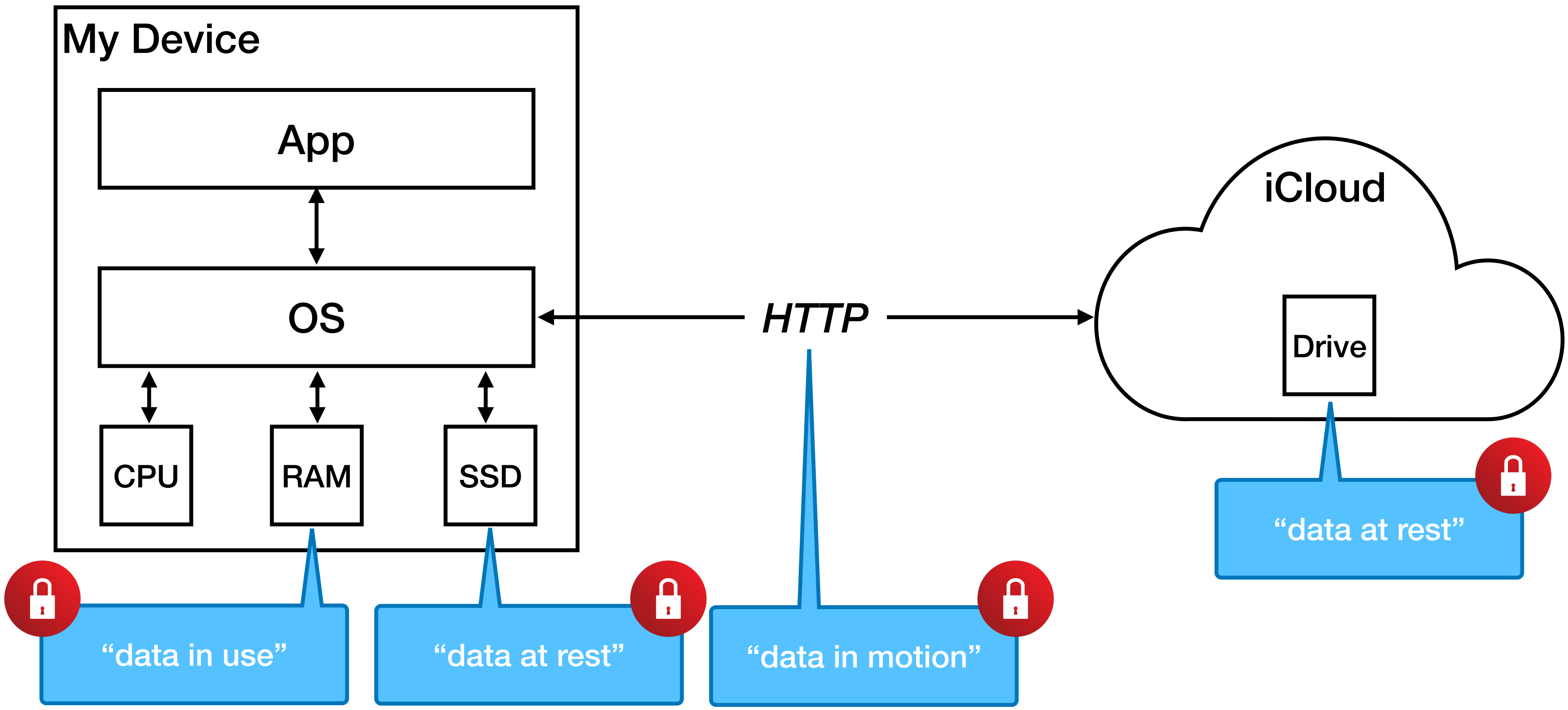
- Bad actors often attempt to
  - decrypt encrypted data without having the encryption key
  - Discover account passwords
- This kind of activity is referred to as “cracking”. There are various techniques & methods bad actors can use to “crack”.
- Cybersecurity experts consistently recommend using strong and unique passwords because they thwart cracking.



# Review: where data resides



# “End to End” Encryption



# Cloud Drive Encryption

| | **Standard (i.e. by default)** | **Advanced** |
|---|---|---|
| **Apple iCloud Drive** | • In Motion<br>• At Rest<br>• Key held by Apple | • End-to-end<br>• Key held by you |
| **Google Drive** | • In Motion<br>• At Rest<br>• Key held by Microsoft | • None |
| **Microsoft OneDrive** | • In Motion<br>• At Rest<br>• Key held by Microsoft | • OneDrive includes a feature called "Personal Vault"<br>  • Personal Vault does not appear to provide any greater degree of security than OneDrive's standard features |

# Authentication

## Signing in to an account

- Authentication is the process of **verifying a user's or system's identity**. It ensures that an individual or entity is who they claim to be before granting access to a system or data. Authentication is crucial for **safeguarding sensitive information** and **maintaining the integrity** of online services.

- Authentication is a cornerstone of digital security, serving as the **first line of defense** against unauthorized access to sensitive information and systems.

- Authentication serves several key purposes:

  1. **Security**: By verifying identities, authentication prevents unauthorized access to systems and data.

  2. **Trust**: Secure authentication processes build trust between users and service providers.

  3. **Efficiency**: Streamlined authentication processes can enhance user experience and operational efficiency.

# Authentication: Factors

Authentication factors verify an individual's identity against a pre-established digital record. The primary types of authentication factors are:

1. **Something You Know**: This involves knowledge-based credentials such as passwords, PINs, and security questions. It relies on information that should be known only to the user.

2. **Something You Have**: This factor requires possessing a physical item, such as a smart card, mobile device, or hardware token. It verifies identity through the presence of an object the user knows to possess.

3. **Something You Are**: This factor uses biometric data like fingerprints, facial recognition, or voice patterns to confirm identity. It leverages unique biological traits that are difficult to replicate.

*Ref: cybersecuritynews.com*

# Authentication: Types


- Basic (a.k.a. single-factor) authentication
  *Single-factor authentication requires only one piece of evidence to verify identity, typically a **username and password**. While simple, SFA is vulnerable to attacks such as phishing and brute force due to its reliance on a single factor.*

- Two-Factor authentication (2FA)
  *Two-factor authentication adds a layer of security by requiring two separate factors for verification. Common combinations include a password (something you know) and a temporary code sent to a mobile device (something you have).*

  - Using email or text

  - Using an authenticator app

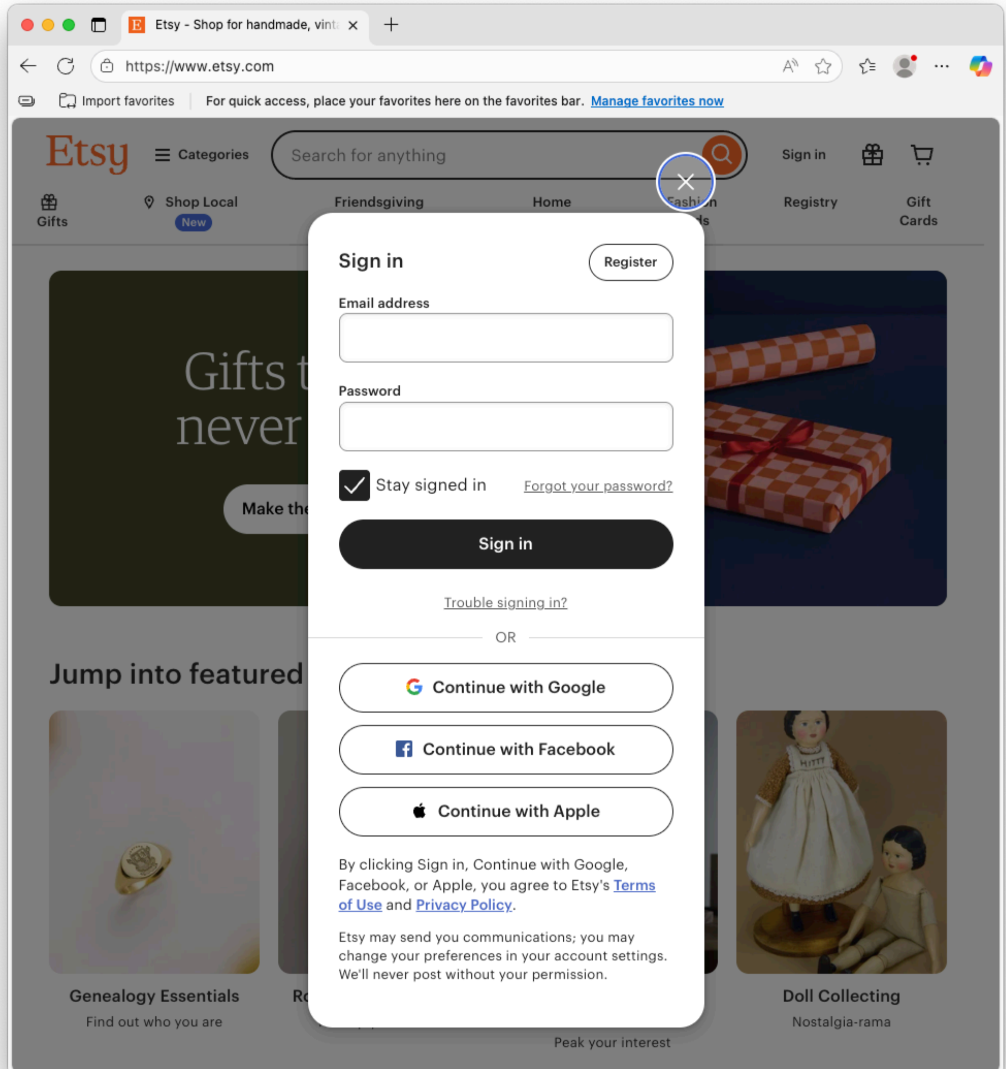
- Multi-Factor (MFA) authentication
  *MFA extends beyond 2FA by incorporating multiple factors from different categories. For example, it might combine a password, a fingerprint scan (something you are), and a smart card (something you have). MFA provides robust security for high-risk environments.*

- Passkeys
  *A method of verifying an app or website user who is tied to both the app or site and the device trying to gain access. Both "keys" need to fit before a user is allowed in, but the process is done without entering a username or other proof of identification.*

# Social Sign-In

Social sign in is an authentication method that lets you log in to a site using using your social accounts on platforms like Google, Apple, Facebook, or LinkedIn. Instead of creating yet another password, you click something like “Sign in with Facebook” or “Continue with Google,” approve a permissions request, and you’re in.

A.k.a., social login, social sign-on, social authentication, or social single sign on (SSO)



# Social Sign-in: Pros & Cons

## Pros

- Easy to use
  *Instead of having to go through the hassle of filling out yet another form with your name, surname, phone number or email address, you simply click on your preferred SSO option and share those (but possibly also other) details with the new app or website. Importantly, your password is never shared with the website – instead, your identity is verified via an authentication token.*

- Reduced “password fatigue”
  *Setting up a strong password with just one of the big internet platforms can give you access to hundreds of other websites, vastly reducing the number of passwords you need to create and memorize.*

## Cons

- Reduced privacy
  *The social site you use to sign in sends your personal info to the site you are signing in to.*

- Single-point of failure

  - *If something goes wrong with the social site you use, you won’t be able to sign in to other sites you sign in to with social sign in.*

  - *If the social site account gets hacked, you also loose control of the other sites you sign in to with the social site.*

# Characteristics of *strong* passwords

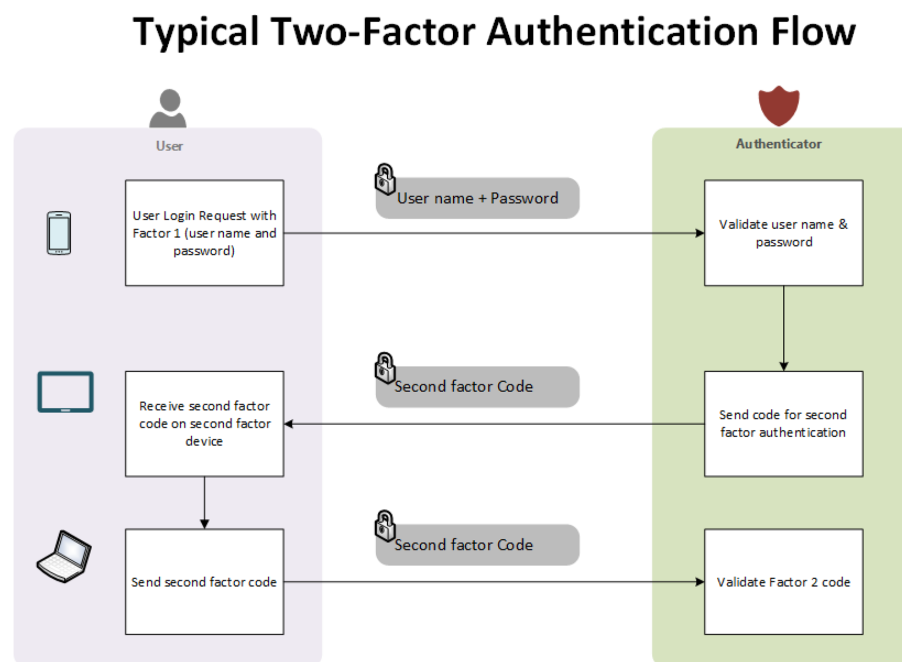
- At least 12 characters long or more
- Combination of uppercase and lowercase letters, numbers, and symbols
- Not a familiar name, person, character, or product
- Is not based on your personal information
- Passwords are unique for each account you have
- Significantly different from your previously used passwords

# Two Factor Authentication (aka, 2FA)

## Also known as Multi-Factor Authentication (MFA)

- Definition: Two-factor authentication is an authentication method in which you are granted access to a website or application only after successfully presenting two or more distinct types of evidence (or factors) to an authentication mechanism. 2FA protects personal data—which may include personal identification or financial assets—from being accessed by an unauthorized third party that may have been able to discover, for example, a single password.

- Accounts with 2FA enabled are significantly less likely to be compromised.



# Authenticator Apps

An authenticator app is a secure and easy identity verification method that generates number codes you enter alongside your credentials to access an account.

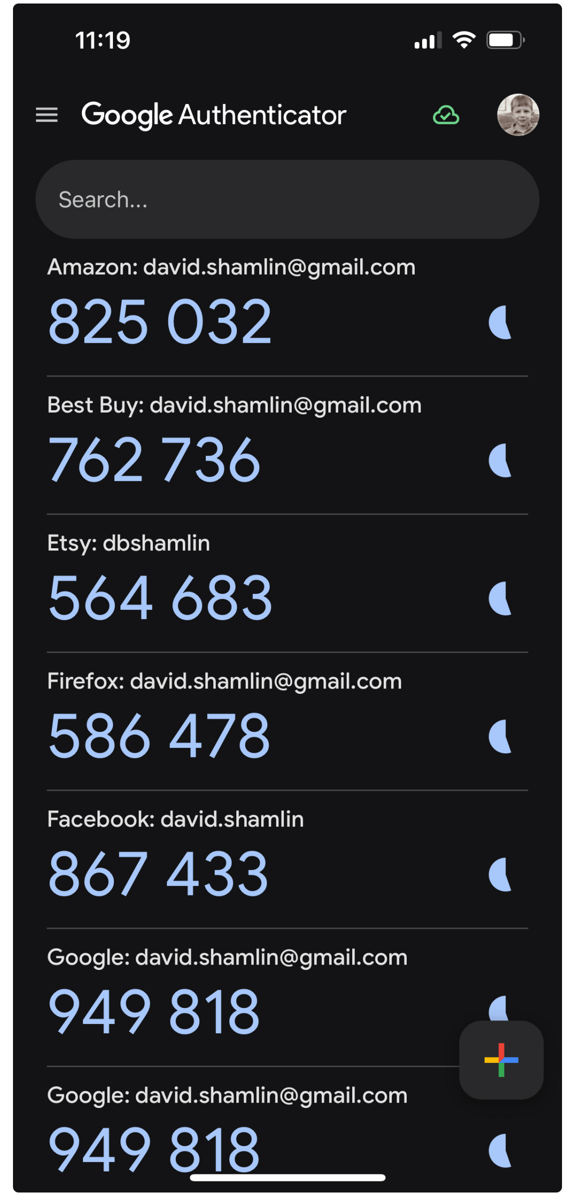
Authenticator apps keep the code local to your device and use encryption to protect the stored secret key. This means the codes aren't transmitted over the internet, which makes them resistant to common attack methods like phishing, SIM swapping and Man-in-the-Middle (MITM) attacks. Additionally, since the codes reset every thirty to sixty seconds, it's extremely difficult for cybercriminals to steal or reuse them.

Note: Not all sights that support 2FA allow you to use an authenticator app; some only allow you to perform 2FA with texts and/or emails.

**Recommended Authenticator Apps**

- Google Authenticator
- Microsoft Authenticator
- Duo
- Bitwarden
- 2FAS
- Authy

**Note**: You can find all of the above in the App Store **on your iPhone.**

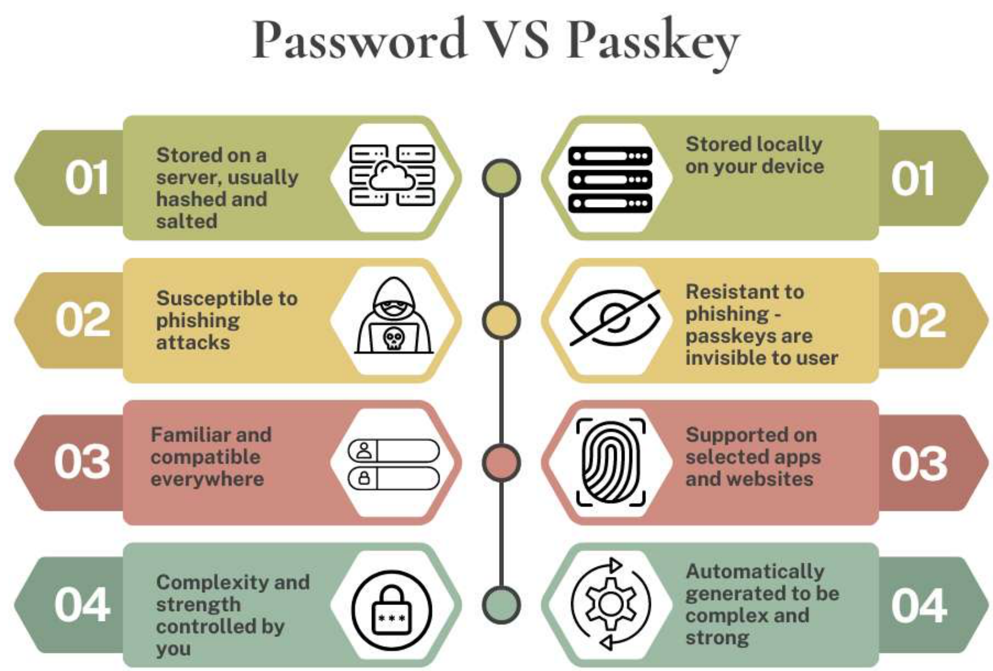


# Passkeys

- **Definition**: Passkeys are digital credentials that use a device like a phone, laptop, or tablet to authenticate users' login attempts. They're a more secure and convenient alternative to traditional credentials, consisting of a username and a password, and are generally considered "phishing resistant" (i.e., extremely difficult—some experts say impossible—for a hacker to steal).

Passkeys came on the scene in 2022. The tech industry wants passkeys to become the standard way we sign in to sites. If that happens it will be at some point in the unforeseeable future as account-based sites (i.e., sites you sign in to) have to be modified to support passkeys.

You need a password to use passkeys due to the way they were designed.



# How Passkeys Work

# Password Managers

- **Definition**: A software application that helps users create, store, and manage their passwords securely. It typically requires a single master password to access an encrypted vault where all other passwords are stored, making it easier to use strong and unique passwords for different accounts.
- Apple's built in password manager is called "Passwords"
  Windows' built in password manager "Credential Manager"
- Some browsers have built-in password managers. **Use with caution!**
  - Passwords stored by a browser's password manager are only accessible when using that browser.
  - Some browser password managers are not sufficiently secure

# Reputable Password Managers

- Bitwarden
  bitwarden.com
- Proton Pass
  proton.me/pass
- 1Password
  1password.com
- Dashlane
  www.dashlane.com/personal-password-manager
- Keeper
  www.keepersecurity.com/personal.html
- Nordpass
  nordpass.com
- LastPass
  lastpass.com

## Password Generators

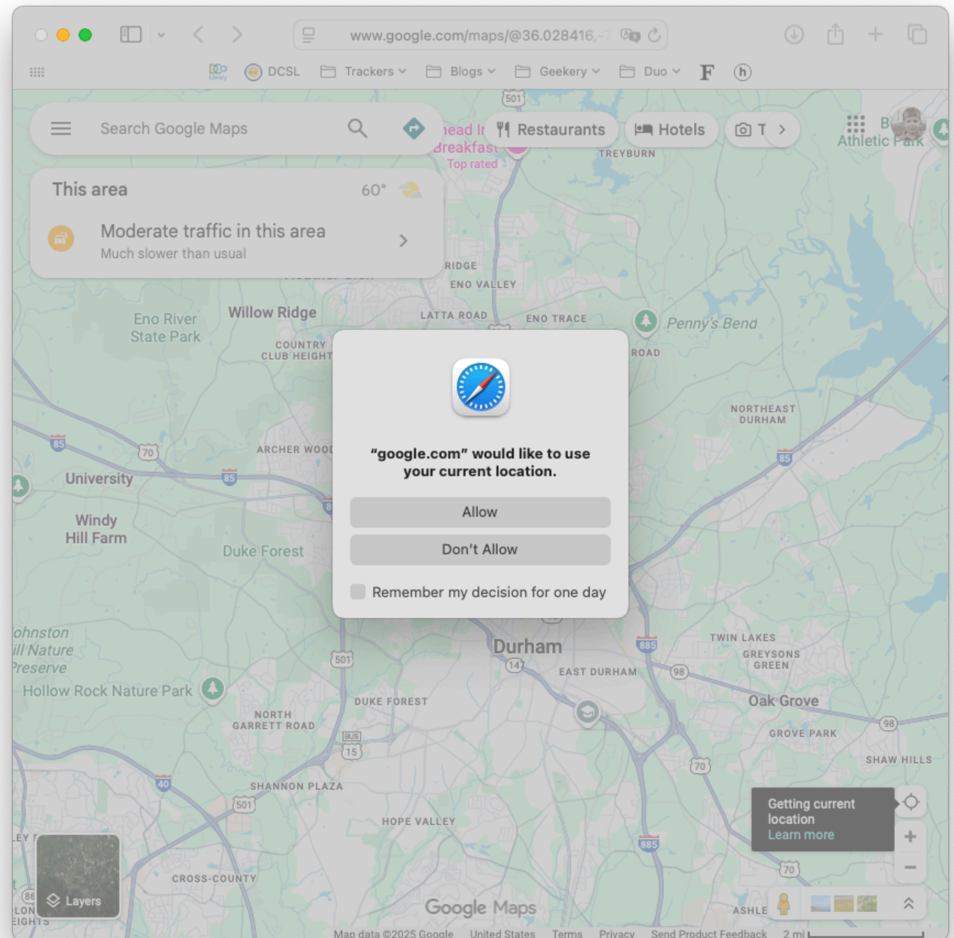
A password generator is a tool that generates a random combination of numbers, letters, and symbols to be used as a password. Password generators let you exclude or include certain characters, allowing you to customize the password complexity to your needs.

Password generators are a fast and convenient way to generate secure passwords. Password managers typically include built-in password generators so you can create a secure password on the fly when signing up for a new account (or updating an old one).

Some password generators allow you to create passphrases, which are passwords that use words instead of fully random characters. Passphrases are easier to remember while still being very secure, assuming they're long enough.

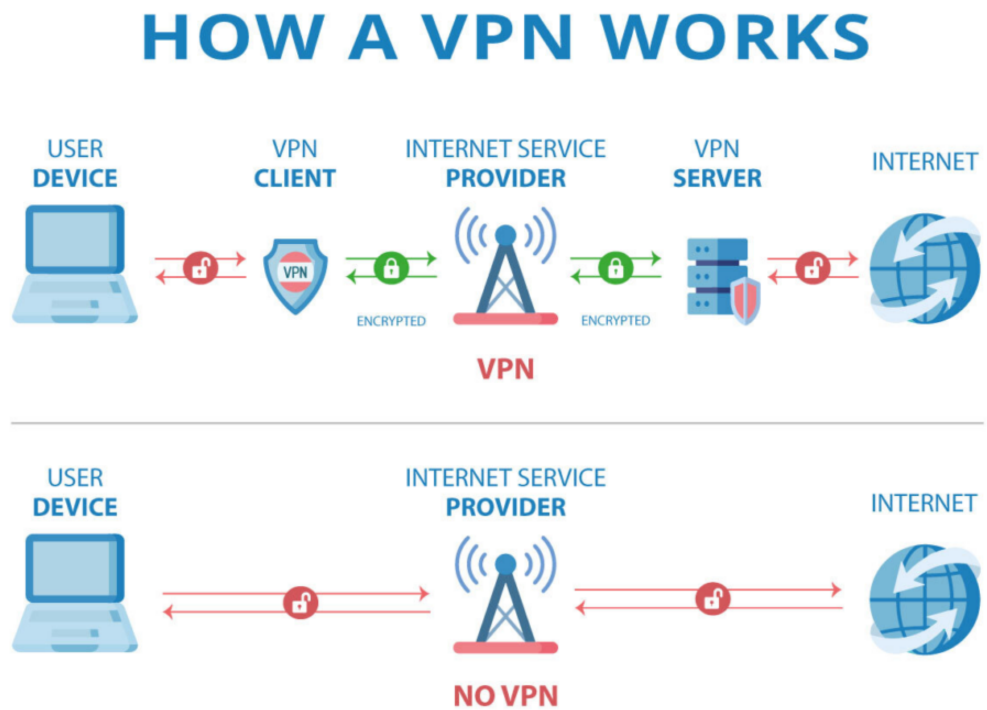
# Authorization

- **Definition:** Giving an app or person access to something
- Examples
  - Allowing a site to know your location
  - Allowing an app like Zoom to access your camera
  - Granting a friend/colleague access to data/files you have on cloud drive



# Virtual Private Network (VPN)

**Definition**: A virtual private network (VPN) encrypts your internet traffic and then routes it through your VPN provider's server before you connect to a website or another online service. This helps disguise your identity and activity on the internet. VPNs can be used to bypass geographic restrictions, shield your activity on public Wi-Fi, and hide your real IP address when you are online.



# Recommended VPN Providers

- ExpressVPN (expressvpn.com)
- NordVPN (nordvpn.com)
- Surfshark (surfshark.com/vpn)
- Proton VPN (protonvpn.com)
- Mullvad VPN (mullvad.net/en/vpn)
- TunnelBear (tunnelbear.com)
- Norton Secure VPN (us.norton.com/products/norton-vpn)
- CyberGhost (cyberghostvpn.com)

# What is Malware?

Malware (short for “malicious software”) refers to software intentionally designed to cause disruption to computers, gain access to (i.e., steal) information you have stored on your computer, and deprive you of access to your data. There is also a category of malware that gives bad actors remote access to your device and allows them to use it as if it was physically in their hands.

Cybersecurity experts categorize malware into sub-categories like

- Viruses: malware that “injects” itself into an app installed on your device in order to do further harm
- Worms: malware that can be installed and run independently of any other apps you have installed; worms can be written to do similar kinds of harm as viruses
- Trojan horses (or simply “trojan”): malware that disguises itself as a normal app. Trojans are often used to facilitate scams and ransomware attacks
- Ransomeware: malware that encrypts the victim’s data until ransom is paid
- Spyware: malware that gathers information from your device and sends it to the bad actor(s)

# How devices get infected with malware

- Downloading software from a site on the Internet (as opposed to through the App Store)
- Pirating software, music, or movies using a file distribution network (e.g., BitTorrent)
- Opening email attachments
- Connecting a disk drive or thumb drive
- Visiting malicious / compromised websites

# Antivirus software

Antivirus software (a.k.a, anti-malware) is an app used to prevent, detect, and remove malware. Antivirus software was originally developed to detect and remove computer viruses, hence the name. However, with the proliferation of other malware, antivirus software started to protect against other computer threats. Some products also include protection from malicious URLs, spam, and phishing.

**Recommended Antivirus Apps**

- Bitdefender Total Security (https://www.bitdefender.com/en-us/consumer/antivirus)
- Norton 360 (https://us.norton.com/products/norton-360-standard)
- McAfee+ Premium (Click here to see more)

macOS has and Windows both include antivirus software. Many cybersecurity experts consider the capabilities of these built-in antivirus features sufficient malware protection. iOS does **not** include antivirus software; some other built-in security features along with the way Apple manages and constrains the distribution of iOS apps through the Apple App Store are sufficient to prevent iPhones and iPads from becoming infected—we may see iPhones and iPads become more vulnerable to malware in the future as hackers become more sophisticated with their attacks.

# Review: Why cybersecurity is important

- What am I protecting: digital assets
  - Accounts (i.e., IDs & passwords, passkeys)
  - Information (i.e., data) stored online (e.g., files/data in cloud storage)
- What am I preventing:
  - Loosing control of access to accounts and/or your data
  - Falling victim to theft and/or fraud
  - Device infection
- Who am I protecting myself from: hackers / hacker gangs, “bad actors”, “malicious agents”, “threat actors”, “cyber criminals”, “cyber terrorists”, scammers

# Review: Security Best Practices

- Use **strong** passwords
- Use **unique** passwords
- Use **two-factor authentication** (2FA) a.k.a. multi-factor authentication (MFA)
- Routinely change your passwords

- Use **passkeys**
- Use a **password manager**

- Use **VPN**
- Use **antivirus** software

- **Encrypt** *confidential* data you store *in the cloud* **or avoid storing confidential data in the cloud**
- Keep software (apps & OS) **up to date**

Everyone should…

- Configure your devices so they automatically update OS and apps
- Always and only use strong and unique passwords
- Use 2FA when the site supports it
- Change an account's password if/when you learn the associated site has been "breached" or "hacked"
- Only store data in the cloud that you really need keep in sync across your devices

Once you've mastered the green items…

- Pick a password manager; learn how to use it
- Try converting a few of your accounts from password/2FA to passkeys

Some people should…

- Use **VPN** if you use the WiFi outside your home
- Use **antivirus software** if you install software you downloaded from a place other than the App Store